

System Level Design and Modelling

COL 812: II Semester 2025-26

Slot M: Mon/Thu 5:00-6:20 PM

LH 603

Course Outline

Instructor: Preeti Ranjan Panda

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Topics

- System Specification: Models and Languages
- Hardware/Software Co-design and Partitioning
- Customising programmable components
 - Processors
 - Hardware Accelerators
 - Memory Architectures and Management
- Power/Energy/Thermal Optimisation and Estimation
- Exploratory research topics

Course Content

- The course will address issues arising in the specification, design, and implementation of complex Systems-on-Chip (SoC) consisting of Hardware and Software components
- Among the questions addressed, will be:
 - How to specify complex systems-on-chip? Some models of computation and implementation languages (SystemC).
 - Which parts of a system should be implemented in software and which in hardware? The Hardware/Software partitioning problem.
 - How to customize and tune processor and memory in an Application-specific way?
 - How do we introduce power/energy/thermal awareness into SoC design?
- Apart from the above topics, exploratory projects will be assigned. It is assumed that the students will make original contributions to open research problems.

Necessary Background

- Digital Logic and Computer Architecture
- Data Structures and Programming

Grading

- Mid-semester Exam: 30%
- End-semester Exam: 30%
- Project: 40%